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VideoDrift: Plug-and-Play Video Refinement for Diffusion Models via KV-Anchored Attention

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Figure 1. **VideoDrift for Video Generation Refinement.** We propose ***VideoDrift***, a post-processing technique that enhances the perceptual quality of videos generated by diffusion models with minimal inference overhead. Given the videos on the top row generated by different video generators, VideoDrift produced videos with enhanced quality in the bottom row.

Abstract

*Text-to-video diffusion models frequently produce videos with degraded perceptual quality, including blurry frames, misalignment with the input text, implausible details, and poor color or structural fidelity, which reduces realism and overall visual quality. Prior video generation enhancement methods often require re-running or modifying the video diffusion sampler, e.g., extra sampling iterations, in-loop optimization, or changes to sampler internals, which is computationally heavy and sometimes unavailable in practical deployments. We propose **VideoDrift**, a train-once, post-generation, frame-wise enhancer that operates directly on rendered videos, leaving the base model untouched. At a high level, VideoDrift inverts each frame and re-renders it with a one-step strong image diffusion model, while a minimal temporal anchoring mechanism stabilizes appearance across frames. Therefore no re-sampling of the original video model is needed. The result is a plug-and-play, backbone-agnostic enhancer with low compute: each frame requires only inversion plus one generator pass, making it easy to deploy on outputs from diverse T2V systems. Through extensive evaluations on VBench, we demonstrate that VideoDrift improves video aesthetics/quality, as well as higher overall scores, while remaining plug-and-play and computationally light.*

1. Introduction

Diffusion models have revolutionized the field of generative vision models, demonstrating impressive text-to-image (T2I) generation quality [10, 25]. Recently, diffusion-based text-to-video (T2V) systems [2, 9, 14, 32] have carried that strength into the temporal domain, yielding compelling motion and scene dynamics. Despite the impressive results of commercial T2V systems [17, 28], open-source counterparts [2, 9, 14, 32] are often limited in fidelity, resulting in distorted figures, blurry visuals, missing details, and inconsistent content.

Many efforts have been proposed to address this problem. To enhance video perceptual quality, prior efforts focused on steering the video sampling process itself. FreeInit [29] repeatedly refines the low-frequency components of the initial noise to bridge a training-inference mismatch. It is training-free but computationally intensive: for every refinement iteration, it reruns the entire generation process. VideoGuide [13] injects a teacher model's denoised samples into the student's early steps, improving temporal behavior without training but still requiring access to both samplers and additional steps. FlowMo [19] optimizes temporal variance of the model's own latent predictions via gradient steps at selected timesteps, yielding better motion coherence at the cost of slower inference. Nevertheless, these methods share a key limitation: they require optimization during the

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video denoising process, making them **slow** and **computationally intensive**, and impractical for real-world use.

In this paper, we propose a fundamentally different approach to video enhancement. We reframe the task as a lightweight, post-processing task that leverages the initial video output as a rich prior, thereby decoupling enhancement from the computational burden of generation. To achieve this, we present **VideoDrift**, a highly efficient framework that enhances each frame independently in a single, non-iterative step, eliminating the need for costly optimization during inference. Specifically, VideoDrift operates by re-rendering each frame with a powerful, one-step image diffusion model, which effectively "repaints" the frame with greater detail and clarity. Specifically, given a video frames, we invert them to inverted noise, then use a robust 1-step T2I model to re-generate it. This design offers several key advantages for VideoDrift: (1) it leverages the **high fidelity and stability** of modern T2I models, which outperform current video counterparts, particularly in the one-step regime; (2) it preserves the **lightweight, cost-efficient** architecture of T2I models; and (3) it inherits their **real-time performance** with minimal latency.

To implement VideoDrift, we address two main challenges. First, re-rendering each video frame with a one-step image generator requires inverting the frame back into the generator's initial noise latent. Standard diffusion inversion techniques [24] are iterative, slow, and not directly compatible with one-step T2I models. Inspired by prior work SwiftEdit [16], we train a lightweight inversion network that maps each input frame directly to its corresponding noise in a single forward pass. In original SwiftEdit, they adopt a two-stage strategy: (1) pretrain the inversion network on synthetic images generated by the one-step model, and (2) fine-tune the pretrained model on real images to enhance realism and generalization. However, naively applying their objectives during Stage 2 leads to blurry reconstructions with noticeable artifacts. To address this, we employ an **additional adversarial loss** to enforce stronger realism and sharper visual quality.

Second, frame-by-frame enhancement ignores the temporal and visual consistency across frames. To address this limitation, we introduce **KV-Anchored Inference**. This mechanism selects one or more frames as anchors and caches their corresponding Key-Value (K-V) pairs from the self-attention modules. When enhancing video frames, we concatenate the anchored K-V features with the current enhanced frame's features. Therefore, the current frame can reference the anchor(s) to produce a temporally consistent and visually stable enhancement.

With our proposals, VideoDrift operates as a lightweight, plug-and-play enhancement module that requires only training an inversion model and a frozen one-step T2I model. Our design brings three key practical advantages: (i) **Efficiency**: each frame is enhanced with two forward passes (through inversion network then T2I generator), avoiding any multi-step video diffusion loop, (ii) **Plug-and-play**: it works on outputs of arbitrary T2V backbones without steering their generation process, and (iii) **Quality**: by re-rendering with the strong T2I prior, VideoDrift consistently improves frame-level perceptual metrics on VBench while preserving motion smoothness, achieving higher overall scores compared to both the raw video and FreeInit. Our contributions can be summarized as:

1. We propose **VideoDrift**, a post-generation video enhancement framework that re-renders frames with a frozen one-step image generator via a learned inversion with no access to the base T2V sampler.
2. We extend SwiftEdit's two-stage inversion scheme with **LADD perceptual loss** on real images. This additional objective significantly improves high-frequency detail and aesthetics quality, enable enhancement by noise produced by the inversion network.
3. To maintain temporal consistency without changing model weight, we introduce **KV-Anchored Inference**, a minimal anchoring mechanism. This enforces the enhancement to align with a stable reference, leading to consistent appearance with minimal computation overhead.
4. Extensive evaluation confirms that VideoDrift produces comparative gains in perceptual-quality metrics and higher overall VBench scores, with substantially lower compute than sampling-time baselines.

2. Related Works

2.1. Text-to-image Diffusion Models

Diffusion models have become a cornerstone in the text-to-image generation task [10, 24, 30], generating high-quality images by iteratively denoising random noise. Latent Diffusion Models [18] improved efficiency by operating in a compressed latent space instead of pixel space, and UNet-based latent diffusion enabled high-resolution image synthesis. Recent work replaces the U-Net backbone with transformers [26] that operate directly on latent patch tokens. PixArt [4, 5] shows that DiT backbones can be trained quickly while producing high-quality, high-resolution images. Stable Diffusion 3 [8] and FLUX [12] scale DiT with Multimodal Diffusion Transformers (MM-DiTs), which keep separate image/text branches and perform bidirectional mixing, improving text understanding and typography. SANA [6, 7, 30, 31] introduces a deep-compression autoencoder and linear attention, enabling strong text-image alignment and fast sampling.

2.2. Text-to-video Diffusion Models

Extending diffusion to video generation requires modeling temporal coherence in addition to spatial fidelity. Animate-

Diff [9] augments a frozen T2I model with trainable temporal attention blocks to capture frame-to-frame correlations. Other works train video diffusion models end-to-end on video data: CogVideo [11] and CogVideoX [32] employ large-scale diffusion transformers with 3D video autoencoders to generate longer clips aligned with textual narratives, and open-source frameworks like VideoCrafter [2, 3] achieve cinematic quality by fine-tuning on video while leveraging pretrained image priors. Still, these models often produce unrealistic videos with distortions and artifacts, missing or implausible details, as we further illustrate in Sec. 5.

2.3. One-Step T2I Models

Standard diffusion models often require 20–100 denoising steps, making them unsuitable for real-time or interactive applications, so recent work focuses on accelerating sampling. Adversarial Diffusion Distillation (ADD) [20, 21] uses a pretrained multi-step diffusion model as a teacher and trains a student to match the teacher in 1–4 steps, with an adversarial loss for sharper details. Score distillation is another successful direction: SwiftBrush and SwiftBrush v2 [15] introduces image-free distillation via Variational Score Distillation (VSD) [27] to train a one-step T2I generator without training images. DMD2 [33, 34] distills a multi-step teacher into a single/few-step generator with improved distribution matching and adversarial objectives, scaling to megapixel images via SDXL distillation. Given their quality and speed, these one-step T2I models are natural plug-and-play backbones for video refinement: a video enhancer can redraw each frame with guidance, obtaining the benefits of a full diffusion model without large per-frame latency.

2.4. Video Refinement Methods

To mitigate temporal inconsistency and quality degradation in T2V models, several video refinement methods aim to improve motion smoothness, preserve subject identity across frames, and reduce flicker. FreeInit [29] is an inference-time strategy that addresses the initialization gap between training and generation by iteratively refining the low-frequency components of the initial latent noise across video frames. FlowMo [19] is a training-free guidance method that steers the denoising trajectory using the model's own intermediate latents and patch-wise temporal variance. VideoGuide [13] leverages a pretrained video diffusion model as a temporal guide for another base model during sampling, while ByTheWay [1] operates directly on temporal attention blocks to improve patch-level consistency. However, these approaches require access to and modification of the internal sampling process and introduce substantial extra sampling cost, making them less suitable for lightweight, post-hoc enhancement of generated videos.

3. Preliminaries

3.1. Diffusion Model

Denoising diffusion probabilistic models (DDPMs) [10, 18] generate data by reversing a forward noising process. Starting from Gaussian noise, a neural network ϵ_θ iteratively denoises the latent over T steps to produce an image x_0. Formally, the forward process constructs a Markov chain:

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^{T} \mathcal{N}\big(x_t;\ \sqrt{1-\beta_t}x_{t-1},\ \beta_t \mathbf{I}\big),$$

where $x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \varepsilon$, $\varepsilon \sim \mathcal{N}(0, \mathbf{I})$, $\alpha_t = 1-\beta_t$, $\bar{\alpha}_t = \prod_{k=1}^{t} \alpha_k$. During inference, the network ϵ_θ predicts either the score $\nabla_x \log q_t(x)$ or the mean of $p_\theta(x_{t-1} \mid x_t)$ to iteratively denoise from an initial Gaussian noise $x_T \sim \mathcal{N}(0, 1)$. Continuous-time models formulate the forward process as a stochastic differential equation (SDE):

$$\mathrm{d}x = f(x,t)\,\mathrm{d}t + g(t)\,\mathrm{d}w,$$

with the corresponding reverse-time SDE

$$\mathrm{d}x = \big[f(x,t) - g(t)^2 \nabla_x \log q_t(x)\big]\,\mathrm{d}t + g(t)\,\mathrm{d}\bar{w}.$$

While early models operate in pixel space, latent diffusion models [18] map data into a pretrained VAE compressed latent z, improving efficiency.

3.2. Attention Mechanism

Attention mechanisms [18, 26] are integral for capturing long-range dependencies, allowing a model to determine *which features interact* beyond local convolutions. In each attention block, input features are linearly projected to form queries ($\mathbf{Q}$), keys ($\mathbf{K}$), and values ($\mathbf{V}$). The output is computed via scaled dot-product attention:

$$\mathrm{Attn}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \mathrm{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^\top}{\sqrt{d}}\right)\mathbf{V},$$

where d is the dimensionality of hidden features. This mechanism is typically extended to multiple heads, enabling the model to jointly attend to information from different representation subspaces.

In T2I models, two attention types are commonly used: **cross-attention** and **self-attention**. Cross-attention layers incorporate a conditional input c, e.g., the text prompt embedding, therefore aligning the network features with the conditional input and enforcing the output to follow the condition faithfully. Self-attention layers instead let the features attend to themselves, connecting distant pixels within a frame based on their correlation, thus improving global structure and identity.

While self-attention is typically designed to connect pixels within an image, it can be extended to connect pixels

across different images when they are concatenated, as applied in multi-view generation [23], enforcing consistency between generated views. Note that when computing self-attention, concatenating images or concatenating keys and values have the same effect. Based on this insight, we develop our frame-anchoring mechanism to enforce consistent enhancements across frames, as described in Sec. 4.3

4. Method

In this section, we propose **VideoDrift**, a practical and effective solution to handle the synthesized video enhancement problem. First, we present an overview of our method and our motivations in Sec. 4.1. Then we discuss how to handle technical challenges of our framework, including robust inversion network training (Sec. 4.2) and temporal consistency enforcement with a K-V anchoring mechanism (Sec. 4.3). Our inference scheme is illustrated in Fig. 2.

4.1. Problem Overview and Motivation

As discussed in Sec. 1, most existing video enhancement methods operate inside the text-to-video diffusion process. They modify the sampling trajectory, inject additional guidance signals, or alter temporal attention blocks of the base T2V model. While effective, these approaches share two practical limitations. First, they require direct access to the internal weights, latents, or sampler of the underlying video model, making them difficult to apply to black-box systems. Second, they typically add multiple extra diffusion iterations or expensive guidance steps, which substantially increase inference cost and slow down video generation. As a result, they are not suitable for a post-hoc enhancer for generic T2V outputs.

To enable a simple and plug-and-play pipeline, given a generated video, we re-render it frame by frame using a pre-trained 1-step T2I generator f_θ. Specifically, let $x_{\text{gen}}^{1:K} = \{x_{\text{gen}}^k\}_{k=1}^K$ denote a rendered video (pixel space) produced by an arbitrary T2V model from text condition c. Let $\mathcal{E}_{\text{VAE}}$ and $\mathcal{D}_{\text{VAE}}$ be the pretrained VAE encoder/decoder of the target 1-step T2I generator f_θ. We first encode each frame to its correspoding latent:

$$z_0^k = \mathcal{E}_{\text{VAE}}\big(x_{\text{gen}}^k\big), \quad k = 1, \dots, K. \tag{1}$$

A light inversion network E_ϕ maps z_k to a noise latent $\hat{\epsilon}_k$ compatible with f_θ:

$$\hat{\epsilon}^k = E_\phi\big(z_0^k, c\big). \tag{2}$$

The frozen generator then *re-renders* the frame in one step:

$$\hat{z}_0^k = f_\theta\big(\hat{\epsilon}^k, c\big), \quad \hat{x}^k = \mathcal{D}_{\text{VAE}}\big(\hat{z}_0^k\big). \tag{3}$$

With no changes to the sampler internals and only two additional one-step one-step diffusing model passes, this pipeline makes the enhancement process much more efficient than previous approaches. Since one-step T2I models encode a strong image prior, they can regenerate frames with sharp edges, clean contrast, and high-frequency details. Still, the quality of the regenerated frames heavily depends on the inverted noise produced by E_ϕ. How to train E_ϕ that can perform inversion in one step and produce "good" inverted noises that improve quality rather than merely reconstruct the original inputs is a challenging problem. Another issue is how to preserve temporal consistency when frames are redrawn independently. These problems are critical, deciding the success of our proposed framework. We will address these problems in the following sections.

4.2. Enhanced One-Step Inversion Network Training

The inversion module E_ϕ is critical; it provides the noise input to the 1-step T2I model for repainting each video frame. However, inverting a one-step model is not trivial; traditional methods like DDIM Inversion are designed for multi-step models, and would fail miserably. A recent work, SwiftEdit [16], suggests that we can train a one-step inversion network to handle this task effectively. The inversion network uses the same backbone and is initialized from the 1-step T2I model. SwiftEdit leverages a two-stage training that first learns exact inversion on synthetic pairs produced by the same one-step generator, then adapts to real images.

Stage 1 - Warm up. Synthetic inversion on generator outputs. For better convergence, we init inversion network E_ϕ from one-step generator f_θ to take advantage of its rich prior knowledge. In the first training stage, a strong supervision on synthetic data is applied to change the network's behavior from generation to inversion. Specifically, we first sample the input noise $\epsilon \sim \mathcal{N}(0, I)$ and render its image latent produced by the one-step generator $z_0 = f_\theta(\epsilon, c)$. We then pass that synthetic input to the inversion network to predict the inverted noise $\hat{\epsilon} = E_\phi(z_0, c)$ and reconstruct the image latent $\hat{z}_0 = f_\theta(\hat{\epsilon}, c)$. We optimize E_ϕ to minimize the reconstruction losses on both the noise prediction and its reconstructed latent:

$$\mathcal{L}_{S1} = \underbrace{\|\epsilon - \hat{\epsilon}\|_2^2}_{\mathcal{L}_{\text{recon}}^{\epsilon}} + \underbrace{\|z_0 - \hat{z}_0\|_2^2}_{\mathcal{L}_{\text{recon}}^{z}}.$$

Stage 2 - Training. Adaptation on real images. While stage 1 changes the inversion network's behavior to inversion, the network still performs poorly on real images due to the domain gap. Hence, a second training stage for domain adaptation is needed. For each real sample image $x_{data} \sim \mathcal{D}_{\text{real}}$, the system computes its latent $z_{data} = \mathcal{E}_{\text{VAE}}(x_{data})$, predicts the inverted noise $\hat{\epsilon} = E_\phi(z_{data}, c)$, and reconstructs the image latent $\hat{z}_0 = f_\theta(\hat{\epsilon}; c)$. SwiftEdit proposes to

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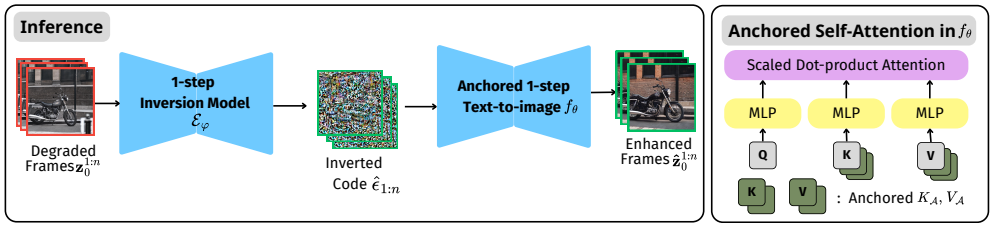


Figure 2. **Inference stage of VideoDrift**. Given a rendered clip, each frame is first inverted by the one-step inversion network E_ϕ, then re-rendered by the frozen one-step T2I generator f_θ. We capture and cache self-attention keys/values from one or more anchor frames by running f_θ once on those anchors. Then when enhancing frames, the cached $K_\mathcal{A}, V_\mathcal{A}$ are concatenated to the current frame's K, V in every enabled self-attention block Anchored Self-Attention.

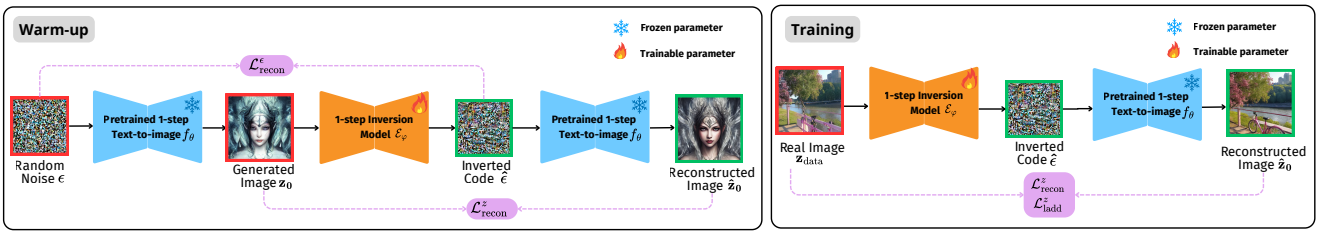


Figure 3. **Inversion network 2-stage training**. Stage 1 (warm-up): train on synthetic pairs from f_θ to predict the exact inverted noise and reconstruct the generator's outputs with L2 losses. Stage 2 (real-image adaptation): encode real images with the VAE, predict inverted noise with E_ϕ, and re-render via f_θ while optimizing a reconstruction loss plus a latent adversarial diffusion distillation (LADD) objective.

finetune E_ϕ using a reconstruction loss on the reconstructed latent and a regularization loss. We empirically found the regularization loss ineffective, thus using only the reconstruction loss:

$$\mathcal{L}_{\text{S2}} = \underbrace{\|z_{data} - \hat{z}_0\|_2^2}_{\mathcal{L}^x_{\text{recon}}} \tag{4}$$

However, with this scheme, the inversion network cannot help to enhance $\hat{z}_0$ from synthetic z_0, as we demonstrated in Sec. 5. We observe that training with just L2 loss gives soft, low-contrast image outputs. The reason is that pixel-wise L2 loss penalizes deviations uniformly, thus preferring to average plausible outputs, leading to blurry outcomes. To improve perceptual quality and robustness in the latent space, we adopt the adversarial loss proposed by LADD [21] into stage 2 training (Fig. 5). Crucially, this loss not only guides E_ϕ to produce inverted noise that reconstructs the real image input more faithfully but also shifts the inverted noise prediction toward the "good" noise manifold of the generator, in which the reconstructed latent $\hat{z}_0$ **resembles the real-image quality**. This domain shift encourages our inversion-regeneration scheme to produce high-quality outputs regardless of the input frame quality; thereby inducing the enhancement effect.

Concretely, LADD loss requires a discriminator D that takes a clean latent, a diffusion timestep, and the text condition as input:

$$D : (z, t, c) \mapsto \text{logits},$$

where z is either a real latent $z_{data} = \mathcal{E}(x_{data})$ or a reconstructed latent $\hat{z}_0 = f_\theta(E_\phi(z_{data}, c), c)$, t is a fixed timestep (e.g., the terminal step used by the one-step generator), and c is the same text embedding as in the inversion network. During training, D learns to discriminate real latents from generated latents, while E_ϕ is updated to fool D by producing latents that are indistinguishable from real ones.

We optimize two coupled objectives. For the discriminator, we use a standard binary cross-entropy loss:

$$\mathcal{L}_D = \mathbb{E}_{z_{data}}\big[\text{BCE}(D(z_{data}, t, c), 1)\big] + \mathbb{E}_{\hat{z}_0}\big[\text{BCE}(D(\hat{z}_0, t, c), 0)\big]$$

For the inversion network (generator side), the LADD term encourages $\hat{z}_0$ to be classified as real:

$$\mathcal{L}_{\text{adv}} = \mathbb{E}_{\hat{z}_0}\big[\text{BCE}(D(\hat{z}_0, t, c), 1)\big].$$

We train E_ϕ and D alternatively with a small number of critic steps per generator update. Intuitively, $\mathcal{L}_{\text{LADD}}$ constrains $\hat{z}_0$ to live on the latent manifold of the distilled diffusion model rather than merely matching. In final, stage 2 loss became:

$$\mathcal{L}_{\text{S2}} = \underbrace{\|z_{data} - \hat{z}_0\|_2^2}_{\mathcal{L}^x_{\text{recon}}} + \underbrace{\lambda_{\text{adv}}\mathcal{L}_{\text{LADD}}(z_{data}, \hat{z}_0)}_{\text{adversarial}}. \tag{5}$$

where λ_{adv} controls the strength of adversarial supervision. By the end of Stage 2, the inversion model can take any image or video frame and produce a latent that the frozen generator will decode into a sharper, more detailed version.

4.3. KV-Anchored Inference

Enhancing frames independently, especially with $\mathcal{L}_{\text{LADD}}$, causes inconsistency between frames. We stabilize appearance by giving the generator *a memory* of one or more anchor frames in original video. Inspired by the observation in

Algorithm 1: Inference with KV-Anchored Attention

Input : Frames $\{x_{gen}^k\}_{k=1}^K$, prompt c, frozen 1-step generator f_θ, trained inversion E_ϕ, anchor set $\mathcal{A}$, set of layers $\mathcal{L}$ where KV concat is enabled.

Output: Enhanced frames $\{\hat{x_0}^k\}_{k=1}^K$.

KVCAPTUREBEGIN (generator, tag): *On the next forward pass, record* detached *per-layer K/V tensors at layers $\mathcal{L}$ into repo[tag].*

KVCAPTUREEND (generator): *Stop recording*

KVUSESOURCES (generator, [tags]): *For subsequent forwards, at each layer $\ell \in \mathcal{L}$, concatenate cached K/V from repo[tags] to current K^k/V^k along the sequence dimension before SDPA.*

VAE Encode all frames $\{x_{gen}{}^k\}_{k=1}^K \to \{z_0{}^k\}_{k=1}^K$

for $k \leftarrow 1$ **to** K **do**

  $\hat{\epsilon}^k \leftarrow E_\phi(z^k)$ // inverted noise

Prepare KV-anchoring

foreach $a \in \mathcal{A}$ **do**

  $k_a \leftarrow$ INDEXFROMANCHOR(a)

  KVCAPTUREBEGIN(f_θ, a)

    $_ \leftarrow f_\theta(\hat{\epsilon}^{k_a}, c)$ // one forward pass to cache K/V at layers $\mathcal{L}$

  KVCAPTUREEND(f_θ)

KVUSESOURCES$(f_\theta, \mathcal{A})$ // enable concat of cached K/V

Enhance frames

for $k \leftarrow 1$ **to** K **do**

  $\hat{z_0}^k \leftarrow f_\theta(\hat{\epsilon}^k, c)$ // self-attn uses $[K_k; K_\mathcal{A}], [V_k; V_\mathcal{A}]$ at layers $\mathcal{L}$

VAE Decode all frames $\{\hat{z}_0^k\}_{k=1}^K \to \{\hat{x}_0^k\}_{k=1}^K$

Sec. 3.2, we tackle this with a **Key/Value-Anchored Attention** mechanism – a plug-and-play inference strategy that enforces consistency across frames by reusing internal attention features from reference anchor frame(s).

KV-Anchored Attention. In each self-attention block we concatenate the anchor's keys/values with the current frame's keys/values before the scaled dot-product attention (SDPA). For a current frame, let $\mathbf{Q}_k, \mathbf{K}_k, \mathbf{V}_k$ be the per-head queries, keys, and values at a self-attention layer, and let $(\mathbf{K}_a, \mathbf{V}_a)$ be cached from an anchor frame. We compute:

$$\mathbf{K}' = \left[\mathbf{K}_k\ \mathbf{K}_a\right], \qquad \mathbf{V}' = \left[\mathbf{V}_k\ \mathbf{V}_a\right],$$

$$\text{Attn} = \text{softmax}\left(\frac{\mathbf{Q}_k \mathbf{K}'^\top}{\sqrt{d}}\right)\mathbf{V}'.$$

This lets the current frame attend to stable anchor features while preserving the original attention pattern. We apply this to **self-attention** blocks only.

By anchoring attention on consistent frame(s), VideoDrift significantly stabilizes subjects and backgrounds over time. For example, if the anchor showed a character with a particular outfit and environment, the subsequent frames, when enhanced, will preserve that outfit's color and the background details more faithfully. This is because the generator can infer the anchor's features to draw from, rather than solely relying on the potentially noisy or drifted current-frame encoding. This method does not require any retraining or modification of the generator's weights. It is a runtime augmentation of the attention mechanism that adds negligible overhead (just concatenating feature maps and some extra key/value memory) and can be turned on or off as needed.

KV-Caching. For each video, we only need to compute $\mathbf{K}_a, \mathbf{V}_a$ from anchor frames once, then re-use them for every frame enhancement. Hence, we enable KV Cache to store that $\mathbf{K}_a, \mathbf{V}_a$.

The pseudo-code of the inference process is shown in Algorithm 1. We run f_θ once on the chosen anchor frames with a capture flag set and store, per enabled layer, the detached tensors $\mathbf{K}_a, \mathbf{V}_a \in \mathbb{R}^{B\times H\times L_a\times D}$. During enhancement, each self-attention uses $[\mathbf{K}_k; \mathbf{K}_a]$ and $[\mathbf{V}_k; \mathbf{V}_a]$ along the sequence dimension. This increases attention length from L to $L{+}L_a$, costs a small runtime and VRAM overhead that we control by caching a few anchors per clip. No generator weights or parameters are changed.

5. Experiments

5.1. Quantitative results

Experimental Setup. Following previous work, we evaluate on the standard VBench [35] and report submetrics for subject consistency, background consistency, motion smoothness, aesthetic quality, imaging quality, dynamic degree, and the average. We test on four T2V generators, including AnimateDiff [9], VideoCrafter [2, 3], T2V-Turbo [14], and CogVideoX-1.3B [32]. We train the inversion network with 2 hours of warm-up on synthetic data and 2 hours of fine-tuning on 100k real images from the LAION dataset [22] on 4 NVIDIA A100 40G GPUs.

Results. We illustrate the results in Sec. 5. We first compare VideoDrift with state-of-the-art baselines, including FreeInit [29], VideoGuide [13], ByTheWay [1]; using the AnimateDiff base T2V generator. We can obserse that AnimateDiff gains improvement in aesthetics (+0.082) and imaging (+0.058), with a healthy rise in dynamic degree (+0.064). A small drop in motion smoothness (-0.030) reflects the fact that per-frame rerendering can slightly perturb

Baseline / Variant	subject consistency	background consist	motion smoothness	aesthetic quality	imaging quality	dynamic degree	AVG
AnimateDiff [9]	0.918	0.960	0.966	0.626	0.652	0.723	0.808
+ FreeInit [29]	0.938	**0.974**	0.974	0.644	0.650	0.708	0.815
+ VideoGuide [13]	**0.941**	0.972	**0.990**	0.601	0.651	0.697	0.809
+ ByTheWay [1]	0.931	0.971	0.959	0.631	0.667	0.701	0.810
+ Ours	0.921	0.958	0.936	**0.708**	**0.710**	**0.787**	**0.837**
T2V-Turbo [14]	0.825	0.943	0.766	0.308	0.490	**0.998**	0.722
+ FreeInit [29]	**0.946**	**0.979**	**0.958**	0.630	0.616	0.550	0.780
+ Ours	0.897	0.952	0.917	**0.706**	**0.711**	**0.998**	**0.864**
VideoCrafter [2]	0.956	0.976	0.965	0.628	0.660	**0.625**	0.802
+ FreeInit [29]	**0.962**	**0.980**	**0.970**	0.631	0.643	0.556	0.790
+ Ours	0.960	0.972	0.952	**0.703**	**0.684**	0.597	**0.811**
CogVideoX 1.3B [32]	0.937	0.966	0.971	0.580	0.569	0.660	0.781
+ FreeInit [29]	**0.951**	**0.975**	**0.977**	0.563	0.538	0.643	0.775
+ Ours	0.934	0.958	0.956	**0.607**	**0.634**	**0.699**	**0.798**

Table 1. **Quantitative Comparisons on VBench**. Across all backbones, VideoDrift improves AVG and consistently lifts aesthetic and imaging scores relative to both raw and FreeInit. Best per metric within each backbone block is bolded.

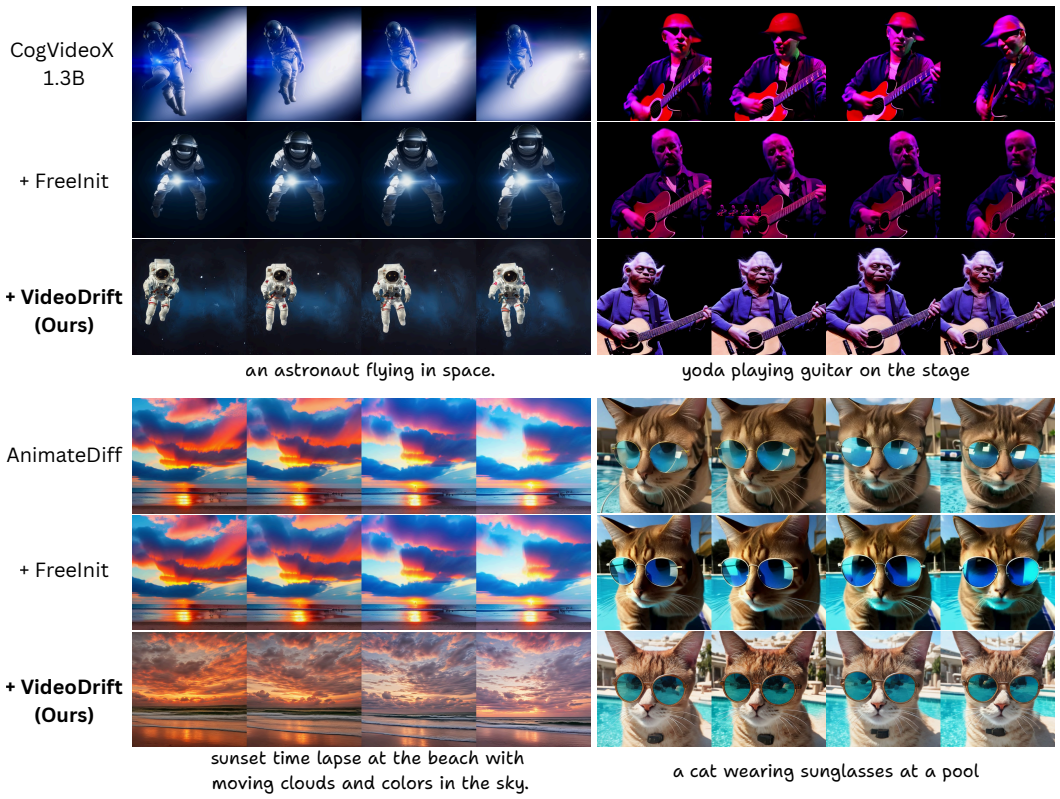


Figure 4. **Qualitative results on CogVideoX 1.3B and AnimateDiff compared to FreeInit**. VideoDrift yields crisper edges, cleaner textures, and more faithful color/contrast while preserving the original motion.

temporal coherence.

We repeat the comparison using the other base T2V generators. Since FreeInit is the best previous video enhancement technique, as verified by the experiments with AnimateDiff, we omit the other baselines in this comparison. As shown in Sec. 5, VideoDrift again outperforms FreeInit on the average scores and improves perceptual quality metrics across all backbones. CogVideoX-1.3B's average score is improved from 0.781 to 0.798, +0.023 over FreeInit. Similar to results with other backbones. The enhancement comes with minor cost in subject/background consistency (-0.003/-0.008) and motion smoothness (-0.015), arising from the per-frame redraw. This cost is permissible given the significant quality benefits and the minimal computation required by VideoDrift. The same scenario happened with T2V-Turbo and VideoCrafter.

Inference speed. We measure compute on AnimateDiff and report the number of function evaluations (NFEs) and generation time per video in Tab. 3. We compare (i) the raw generator, (ii) FreeInit with k refinement iterations (3–5 is typical), and (iii) VideoDrift, which adds only two one-step passes per frame (inversion + generation).

Method	subject_consistency	background_consist	motion_smoothness	aesthetic_quality	imaging_quality	dynamic_degree	AVG
AnimateDiff	0.918	**0.960**	**0.966**	0.626	0.652	0.723	0.808
W/o LADD	0.913	0.956	0.947	0.632	0.651	0.694	0.799
With LADD	**0.921**	0.958	0.936	**0.708**	**0.710**	**0.787**	**0.837**

Table 2. Effect of LADD during inversion training. We keep the VideoDrift setting and remove LADD to see its effect. Adding LADD gains increase in aesthetic and imaging quality, especial in dynamic degree, at a small cost in motion smoothness

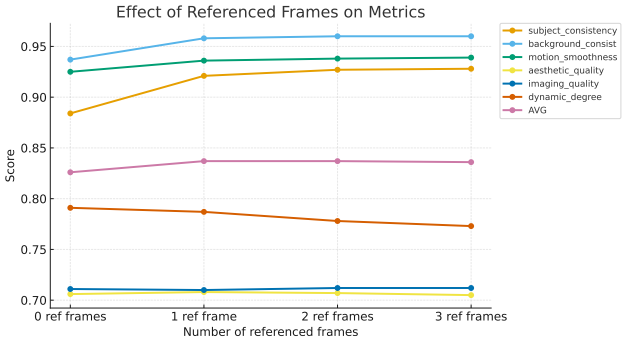


Figure 5. **Effect of the number of anchor frames**. VBench scores when varying referenced anchors from 0 → 3.

Method	NFEs	seconds per sample (s/video)
AnimateDiff	50	41.25
AnimateDiff + FreeInit	50 +50 × $(k-1)$	41.25 +41.25 × $(k-1)$
AnimateDiff + Ours	50 +2	41.25 +3.02

Table 3. **Compute comparison**. We list number of function evaluations (NFEs) and seconds per video. FreeInit re-runs the full sampler k times (50×k NFEs, 41.25×k s). VideoDrift adds only two one-step passes per frame (50+2 NFEs) with 3.02s overhead. kis the number of FreeInit iterations (typically 3–5)

Concretely, with k = 3 to 5 (as commonly used), FreeInit's total runtime becomes 123.75–206.25 s/video, whereas **ours is 44.27 s/video (41.25 + 3.02)**. It is roughly 2.8×–4.7× faster end-to-end than FreeInit for the same backbone and clip length. If we compare just the refinement overhead, our 3.02s is about 27×–55× lower than FreeInit's extra 82.5–165s required for 2–4 additional full passes.

5.2. Qualitative results

Figure 4 illustrates the visual impact of VideoDrift on different T2V backbones. Each triplet shows the raw video, the FreeInit refinement, and our method. Overall, our approach produces sharper frames, more coherent subjects, and visuals that better match the input prompts, while keeping the original motion intact.

For the prompt *"an astronaut flying in space"* with CogVideoX-1.3B, the baseline shows a blurry astronaut on a flat background, and FreeInit mostly just brightens the scene. With VideoDrift, the astronaut's body and helmet are clearer, and in sample *"yoda playing guitar on the stage"*: our method better preserves Yoda's face and outfit, and the guitar looks more consistent with the text.

On AnimateDiff, the *"sunset time lapse at the beach"* example shows that the raw model produces over-saturated, smeared skies. VideoDrift reduces the over-saturation and recovers smoother, more realistic color transitions. In *"a cat wearing sunglasses at a pool"*, our refinement sharpens the glasses' edges and the reflections on the lenses, leading to a cleaner and more detailed appearance than FreeInit.

These examples highlight that VideoDrift consistently improves frame-level perceptual quality: it removes blur and artifacts, corrects extreme color responses, and restores fine details, all while preserving the underlying motion and composition produced by the original T2V models.

5.3. Ablation and Analysis

We ablate the number of anchored frames used by KV-Anchored Attention and the loss terms used in inversion network training. All experiments were done on VBench using AnimateDiff.

Number of Anchored Frames. We test using 0–3 anchor frames. As Sec. 5 shows, an anchor help a lot: subject and background stay more consistent than with none. More anchors like 2 or 3 give similar results while keeping aesthetic/imaging quality high. It also barely matters which frame you pick. Self-attention treats the extra K/V tokens like appearance templates, and within a single clip the look (colors, textures, identity) is stable, so different anchors give nearly the same guidance.

Loss terms. We compare inversion network training with and without the LADD term, keeping the rest of the setup fixed. On AnimateDiff, adding LADD improves the average score by 0.038 points with notable gains in aesthetic (+0.076) and imaging quality (+0.059), and a substantial rise in dynamic degree (+0.093). We observe a small trade-off in motion smoothness (-0.011), which is largely compensated by KV anchoring at inference.

6. Conclusion

In this work, we introduce VideoDrift, an efficient one-step frame-wise enhancer that improves video fidelity while incurring substantially lower computational and time costs than previous approaches. Our method shows significant improvements in frame aesthetics and overall scores on VBench, outperforming state-of-the-art methods while being significantly lighter and faster.

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